

Cosmological Relativity: what the construction delivers, what it declines, and where its edge is

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Abstract

This paper states, in one place, what the Cosmological Relativity corpus establishes, at the weight each result is established at, together with what the construction declines to claim and where its open edge lies. Nothing here is new: every claim is drawn from one of seventeen papers or from the ledgers those papers cite, and is stated at the register its source gives it—a proposition as a proposition, a reading as a reading, a conjecture as a conjecture. What is new is the assembly, and the two things an assembly can supply that a set of papers cannot: the joins, where a result in one paper is load-bearing for a result in another, and the relative size of the frontier against the body of work that surrounds it.

1 The object, and the distinctions it forces

Cosmological Relativity is a construction on one geometric object, and almost everything below is that object read from somewhere. The object is a de Sitter substrate; the reading is a *cut* of it; and the exact solutions of general relativity that the corpus reaches are cuts, not separate geometries. Before any of that can be stated precisely, two things have to be fixed: what kind of object the substrate is, and what a *boundary* of such an object is. The second is not a preliminary. It is the distinction the whole construction rests on, and a reader who does not have it will misread every result that follows.

1.1 Three senses of “real”, fixed before the word is used

The corpus uses *real* in three senses and keeps them apart; this paper does the same. They are not degrees of one thing.

Real in the arithmetic sense: a geometry has a real coordinate basis, with no imaginary coordinate doing any work. That is the sense in which the substrate below is a real manifold, and the claim is about coordinates and signature and nothing more.

Real by construction: a feature is built by a construction and really seen from the vantage that builds it. The perspectival mass is of this kind, and so is the compact face on which a continuous $\mathfrak{su}(3)$ could live.

Existent: the ontological sense, whose criterion in this framework is temporality. *The existent is the evolving three-dimensional layer, and the manifold is its record* [8]. By that criterion the compact face is real by construction but, being Riemannian and atemporal, *not a co-equal existent*—it holds no clock [9].

So nothing in this section claims that a five-dimensional object is what exists. The substrate is the geometric core the construction reads and cuts; what exists, on this framework’s own criterion, is the layer. The higher-dimensional geometry is what the layer is a cut of, not a thing possessing the layer’s kind of being, and the four-dimensional manifold is the record rather than the existent.

1.2 The substrate has a real coordinate basis, and its signature is intrinsic

The maximally symmetric solutions of $R_{\mu\nu} = \Lambda g_{\mu\nu}$ form a one-parameter family with curvature radius $\alpha = \sqrt{3/|\Lambda|}$. They are usually met as hypersurfaces in a flat five-dimensional space, which is a convenient picture and not the manifold: solving that embedding for the fifth coordinate and substituting back returns an intrinsic four-dimensional line element in which the manifold is coordinatised by four *real* coordinates, real regardless of the sign of the curvature. The imaginary fifth coordinate is a property of one embedding picture, not of the surface. *This is reality in the first of the three senses above, and in that one only.*

Reading the metric's eigenvalues off that real line element gives the result the programme's ontology stands on. Three eigenvalues are positive always, and the fourth's sign is fixed by real data alone.

Proposition 1 (The unique intrinsically Lorentzian maximally symmetric manifold). *De Sitter space is the only real maximally symmetric manifold carrying an intrinsic Lorentzian signature. It has a real coordinate basis, satisfies the maximal-symmetry requirement, and is the only such manifold with the requisite causal structure [10, 1].*

Two ways the signature can turn are worth separating, because they are of different kinds. At fixed positive curvature the crossing occurs where the fourth eigenvalue passes through a *pole*: its magnitude diverges and its sign flips, the numerator being constant throughout. *So the metric does not degenerate at the crossing*—a signature change through zero would send the determinant to zero and leave no metric there, and that is not what happens. The one place the eigenvalue vanishes is the null cone: not a crossing within a member of the family, but the family's own singular leaf [10].

That the substrate is reached through imaginary instruments is therefore a fact about the instruments. The embedding coordinate, the equatorial seam's continuation, and the reassignment at the cosmogenesis are each places the construction uses an imaginary variable and lands on a real manifold; the geometry is intrinsically real at every point they land on [10]. In one case the instrument is not optional and the case is the construction's own central algebraic object: over the field of the mass parameter the horizon cubic is irreducible, and below the Nariai mass its discriminant is positive, so its three roots are real and distinct. That is exactly the hypothesis pair of *casus irreducibilis*, whose conclusion is that no root lies in any real radical extension—the *three real horizon radii admit no expression in real radicals as functions of the mass*, and every radical route to them passes through $\mathbb{C}$ [10]. The imaginary is the only available road, and the roots it lands on are real.

1.3 One scale, and what that means

The substrate carries a single dimensionful scale. Read projectively it is a Cayley–Klein geometry^L whose *absolute* is the null cone, and its geodesic separation is the Cayley–Klein log-cross-ratio taken against that absolute, with the Cayley–Klein constant equal to α [10]. So α is not a length inserted into a metric but the constant of a projective geometry whose absolute is the light cone—which is the signature proposition above, read projectively rather than metrically.

The constants through which physics is written on it are unit gauges, each a nameable feature of the substrate: c is the null-ruling slope of the equilateral hyperboloid, Λ the waist, and G enters only as the offset-length GM/c^2 , the mass being the offset of the cut from the central geodesic [10, 6, 4]. On this accounting the gravitational–cosmological–quantum sector spends no free dimensionless constant, and the reason is structural rather than counted: neither real form supplies a second invariant, and a dimensionless magnitude needs two [10].

What the count leaves is not a constant but an initial condition, and the corpus carries it as measured content rather than as a derived quantity—as flat Λ CDM carries the baryon-to-photon ratio from a baryogenesis it does not model [7].

1.4 What a boundary of the substrate is

Here the corpus draws the distinction that everything downstream depends on, and it is a distinction between two kinds of singular locus rather than between a real one and an artefact.

A *metric* singularity is a locus at which a spatial extent contracts to zero while the curvature stays finite. A curvature singularity is a locus at which an invariant of the curvature diverges. The event horizon of a collapsing body is the first kind: a null hypersurface along whose generators the spatial extent also contracts to zero, occurring only as the null future boundary approached at infinite exterior time [2]. Three consequences of that follow immediately, and each is a distinction a reader needs before meeting any application.

Observer-dependence is not the criterion. It is tempting to sort these loci by whether they are seen by every observer, and the sorting fails at once: the Rindler horizon is perspectival and carries a flux, so “perspectival, therefore no flux” is refuted before it starts. What sorts the four horizons the corpus treats is *completion*—whether the boundary is reached on any finite layer—and completion sorts all four [2].

And the finite-curvature case is the one the metric-singularity result treats. It is a different object from the locus standardly called the curvature singularity at $r = 0$, where the invariants computed on the areal radius diverge; the first is established with the event horizon as its exemplar, and nothing is said there about the second beyond noting that it is a different object [2]. *That the two admit a common treatment, and what becomes of the divergence at $r = 0$ under it, is established in the companion* [3, 4]. *No claim of necessity or sufficiency is made here in either direction:* the scope is what is stated, and reading it as a general criterion for when a boundary may be crossed would assert what the source declines to assert.

And formal likeness does not sort the two classes. A chart such as Eddington–Finkelstein parametrises approaches to a singular locus by a coordinate that degenerates there, so a single metric place is drawn spread out as an extended line. Mercator draws the North Pole by the same projective mechanism, and the visual appearance is identical: a place drawn as a line. *But the reason for the appearance is the opposite.* At the pole the chart manufactures an extension that is not there, and a chart centred on the pole collapses the line back to the point it always was. At the horizon the chart spreads out a metric collapse that *is* there: reading the line back yields no ordinary point a better chart would reveal, because no chart removes it—the *collapse is in the metric, not in the projection* [3]. So no argument from how a singular locus is drawn can sort the two, in either direction.

1.5 What follows for the reading, and what does not

Two guards belong here rather than later, because both are misread in the same direction.

Perspectival does not mean unreal. Where the corpus calls a feature perspectival it means built by a construction and really seen from the vantage that builds it—the curvature divergence of a perspectival metric is real as that metric’s, and reading it as an artefact in the pejorative sense is the misreading the epistemology is written to defeat [5, 3].

And identifying a feature as perspectival is not a dismissal of the construction that carries it. Sweeping a radial curve to recover a spatial geometry is the legitimate and indeed the natural way to chart a geometry from a given vantage, and the result agrees with the standard geometry on every observable outside the horizon. What is identified is not an illegitimate operation but a misreading of its product: treating features the construction manufactures—the asymmetric labelling, and with it the curvature singularity at $r = 0$ —as features of the geometry rather than of the chart. *The construction is faithful as a description from its vantage; the error is ontological* [3].

References

- [1] D. Janzen, *A Solution to the Cosmological Problem of Relativity Theory* (PhD dissertation, University of Saskatchewan, 2012), ch. 3.
- [2] D. Janzen, *The causal structure of gravitational collapse: the event horizon as a metric singularity.*
- [3] D. Janzen, *The Schwarzschild circle: one cycloid, two metric singularities.*
- [4] D. Janzen, *The Schwarzschild–de Sitter slicing curve.*
- [5] D. Janzen, *The shadow of existence: inferring the world that casts the appearances.*
- [6] D. Janzen, *The slicing operator: straight cuts are vacuum, matter is the bend of the cut.*
- [7] D. Janzen, *The cosmology of Cosmological Relativity and its scalar perturbation sector.*
- [8] D. Janzen, *Cosmological Relativity: the framework.*
- [9] D. Janzen, *The matter boundary: what the substrate’s symmetry can and cannot carry.*
- [10] D. Janzen, *The geometric core: the substrate as a real object, and maximal symmetry worn seven ways.*

Appendix L The Knowledge Ledgers

Each claim marked ^L in the text rests on a field bake or the figure–theorem bake recorded in the named ledger, which ships with this work and carries the probes, the receipts, and the three registers kept apart — what bit, what bounced, and where the boundary is. A ledger is working apparatus, not a publication, and is cited here rather than in the bibliography. This appendix is generated from the ledger index, not maintained by hand.

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[field bake]

projective geometry of quadrics against the CR substrate — the CK metric identification, and the ladder gap it exposes